

Simultaneous gas-free calibrated fMRI and fNIRS measurements in human visual cortex: a feasibility study

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Impact

The combined gas-free calibrated fMRI and fNIRS approach enhances the physiological interpretability of metabolic mapping, providing a validated basis for multimodal fusion models and future studies of altered neurovascular coupling in patient populations.

Synopsis

Motivation: Gas-free calibrated fMRI is a promising method for measuring brain oxygen metabolism that requires further validation to establish its reliability across cortical regions and subjects.

Goals: Our goal is to explore the relationship between hemodynamic and metabolic estimates exploiting a multimodal framework and to establish a foundation for future fusion models.

Approach: We combined simultaneous gas-free calibrated fMRI and fNIRS during visual stimulation, examining cluster- and channel-wise signals between fMRI (BOLD, CBF, CMRO₂) and fNIRS (HbO, HbR) measurements.

Results: Strong positive relationships emerged, with HbR showing the closest correspondence to CBF (median correlation = 0.8), indicating consistent vascular–metabolic coupling.

Introduction

The calibrated functional magnetic resonance imaging (fMRI) framework offers a quantitative measure cerebral oxygen metabolism¹. Recently, a gas-free calibrated fMRI method² combining blood oxygenation level dependent (BOLD) and arterial spin labelling (ASL) measurements has demonstrated the ability to quantitatively map cerebrovascular function and the cerebral metabolic rate of oxygen (CMRO₂), a key marker of brain energy consumption. Removing the need for respiratory gas challenges makes this approach more easily tolerated and suitable for broader research and clinical applications. However, to interpret variations in metabolic and vascular responses across individuals or populations, we must characterize the intrinsic variability of these measures in healthy subjects, understanding the sources of this variability and potential methodological biases^{3,4}.

Previous work⁵⁻⁷ combining standard hypercapnic calibrated fMRI with functional near-infrared spectroscopy (fNIRS) has highlighted the potential of fNIRS to provide complementary hemodynamic information and translate these protocols into more naturalistic settings. fNIRS offers high temporal resolution and independent quantification of oxy- and deoxyhaemoglobin (HbO, HbR).

In the present study, we assessed the feasibility of simultaneous gas-free calibrated fMRI and fNIRS acquisition over the occipital cortex during visual stimulation. Our goal is to explore the relationship between hemodynamic and metabolic estimates and to establish a foundation for future multimodal fusion models.

Methods

Data acquisition: Concurrent fMRI and continuous-wave fNIRS data were acquired from 10 healthy subjects (males, age: 29 ± 7 years) using a 3T MAGNETOM Prisma scanner (Siemens Healthineers) with a 32-channel receive-only head coil and fNIRS sensors placed over the occipital cortex (NIRx Medical Technologies) (Fig.1A). The imaging protocol (Fig.1B) consisted of an anatomical T1-weighted MP2RAGE⁸ scan; an inversion recovery Gradient-Echo Echo-Planar Imaging (GE-EPI) sequence to estimate haemoglobin concentration from blood T₁ in the superior sagittal sinus⁹; and a pseudo-continuous ASL (pCASL) research application sequence with dual-excitation EPI readout to collect simultaneous BOLD and ASL data¹⁰. Calibration images (M₀) were acquired in blip-up/blip-down modality to correct for field distortion and quantify ASL. Subjects performed a breath-holding (BH) task (gas-free calibration), consisting of 10 cycles of 20s of BH and 40s of recovery. End-tidal traces of CO₂ and O₂ were recorded using a nasal cannula connected to a gas analyser system (ADInstruments). Finally, the subjects underwent a visual stimulation task, consisting of a flickering checkerboard paradigm comprising 10 trials (30s of stimulation+ 30s of rest).

Data processing: MP2RAGE was denoised¹¹, corrected for intensity inhomogeneities (N4biasfieldcorrection, ANTs¹²), segmented (gray and white matter mask from CAT12¹³), and registered into 1-mm isotropic MNI152^{14,15} space (Syn, ANTs). BOLD and ASL images were motion-corrected (AFNI-3dvolreg¹⁶), corrected for field inhomogeneities (FSL-topup¹⁷), and registered to the MP2RAGE (ANTs). BOLD images were despiked (AFNI-3dDespike). The resulting images were used to calculate the M calibration parameter¹⁸. All the functional images were warped to 2-mm isotropic MNI152 space. fNIRS data were pre-processed according to standard guidelines¹⁹. fNIRS and fMRI data were analysed using a first- and second-level General Linear Model block-design framework (SPM²⁰). Activated voxels (FWE $p < 0.05$, cluster-corrected) were used to extract BOLD and CBF time-series, averaged across trials, hemispheres, and subjects. CMRO₂ changes were computed feeding the averaged time-series into the inverse Davis model²¹. For fNIRS, group-level activated channels ($p < 0.01$) were retained and averaged across hemispheres and subjects. Pairwise correlations between fNIRS and fMRI time courses were computed per subject. Inter-modal relationships were assessed considering the 90th percentile of each signal (for fNIRS only channels with $\beta > 0.5\beta_{\max}$). For this analysis and the correlation computations, we used the negative HbR (–HbR) signal.

Results and Discussion

Task-related activations were observed in both modalities (Fig.2), with one cluster per hemisphere and the CBF cluster being more localized. The group-level time-series (Fig.3) showed consistent task-evoked increases in BOLD, CBF, and CMRO₂ in both hemispheres, with corresponding HbO increase and HbR decrease. The fMRI–fNIRS correlations (Fig.4) were generally strong and positive, but with higher inter-subject variability in HbO with a few outliers and wider variance, whereas HbR shows more stable, homogeneous correlations values across subjects (median correlation = 0.8). Linear fits (Fig.5) demonstrated clear positive associations between CBF, BOLD, and CMRO₂, with physiologically coherent coupling. fNIRS–fMRI fits (particularly HbR vs CBF) also showed strong linear trends, though a few points deviate, reflecting inter-subject variability in physiology and data quality. This analysis was restricted to the left hemisphere, as several right-hemisphere channels were pruned in some subjects, making reliable amplitude estimation impossible.

Conclusion

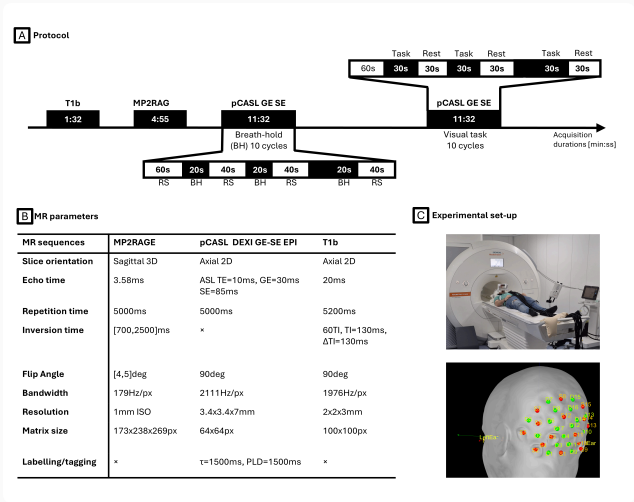
The strong agreement between fNIRS and fMRI responses, particularly with HbR, supports the feasibility of this approach and development of a fusion model. It also highlights the need for developing ad-hoc processing techniques and co-registration methods to handle inter-subject variability.

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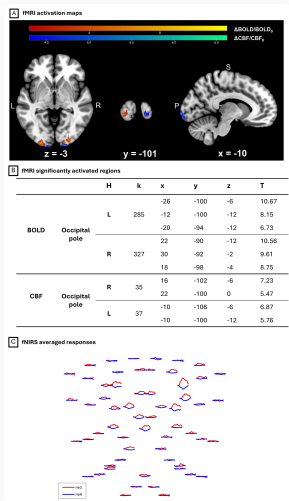
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Figure 1: A: MR protocol. B: list of the parameters for each MR sequence. C: experimental set-up and fNIRS montage on a subject's head.



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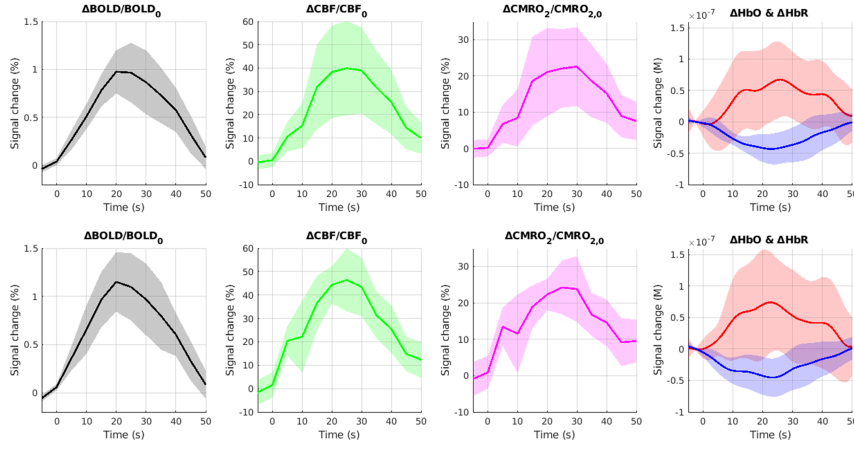
Figure 2: A: group-level fMRI activation maps (t -values) for BOLD and CBF (FWE-corrected $p < 0.05$, cluster level). B: group-level Significantly activated regions for fMRI ($p < 0.05$, FEW-corrected at cluster level). For each cluster, the region with the maximum t -value is listed first, other regions in the cluster are listed below. The hemisphere (H) and the number of voxels (k) are shown for each cluster; the MNI coordinates, the t -value for each peak voxel. C: group-level fNIRS responses for each channel.



LH

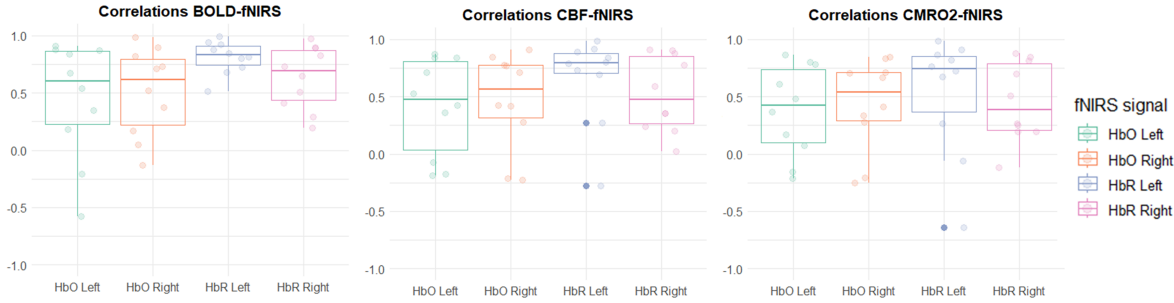


RH



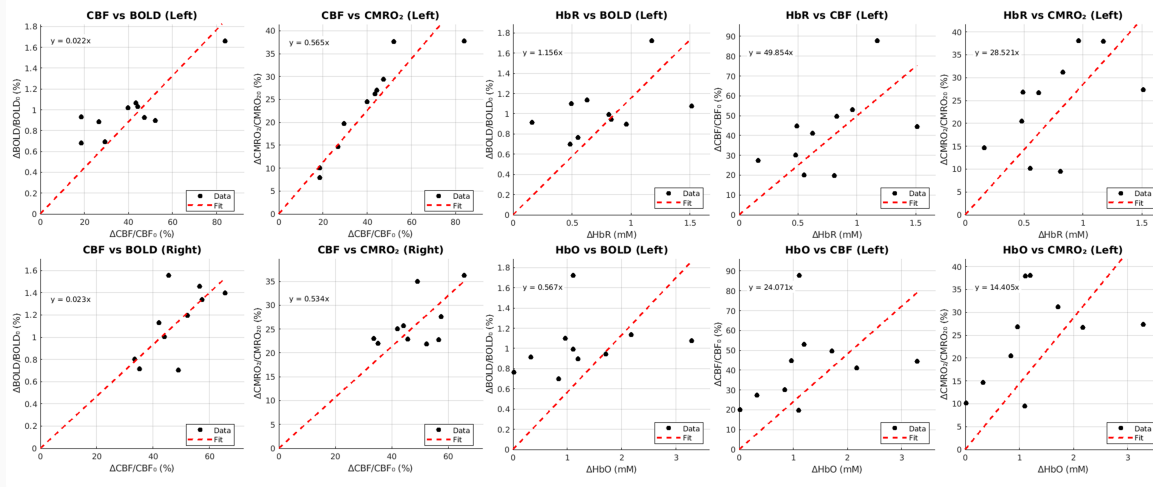
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Figure 3: Group-level averaged time series (bold lines) for each signal and hemisphere. The shaded areas represent their standard deviations. Each row represents the averaged time-series for one hemisphere.



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Figure 4: Boxplots of the pair-wise correlations between fNIRS and fMRI signals for each subject, divided by chromophores and hemispheres.



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Figure 5: Least-square fits (dashed red line) of all the between- and inter-modal measurements. fNIRS signals are presented only for the Left hemisphere, due to heavy channel pruning on the Right one for some subjects.

